



Porous covalent organic frameworks for high transference number polymer-based electrolytes†

Cite this: DOI: 10.1039/c8cc08725c

Received 2nd November 2018,
Accepted 7th January 2019

DOI: 10.1039/c8cc08725c

rsc.li/chemcomm

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While solid polymer electrolytes are poised to be the key component of next-generation solid-state batteries, the low Li^+ transference number of the polymer electrolytes limits their practical applications. Here, porous boron-containing covalent organic frameworks with different surface areas were synthesized and employed as functional additives for enhancing the Li^+ transference number of the polymer electrolytes. The boron-containing frameworks enable strong adsorption of the anions of the lithium salt, leading to a significantly enhanced Li^+ transference number of the polymer electrolyte containing COF additives. It is observed that solid-state cells assembled with the COF-containing polymer electrolytes exhibited remarkably decreased overpotentials and enhanced rate performances, which opens up new ways to apply porous organics in next-generation solid-state batteries.

Solid polymer electrolytes (SPEs) have received considerable attention as a key component in advanced lithium ion batteries (LIBs) due to their high dimensional/thermal stability.¹ Such advantages also enable the usage of lithium metal anodes in next-generation lithium batteries (LBs), which can store large quantities of energy per unit weight and volume.² For the practical application of SPEs, lithium transference number (t_{Li^+}) is an important factor that needs to be considered for fabricating LIBs/LBs with excellent performance. t_{Li^+} is defined as the relative amount of the transporting lithium ion compared to that of the counter anion, which generally has a value in the range of 0.2–0.5 in most common SPEs.³ The low t_{Li^+} of such electrolytes is believed to increase the tendency to form ion concentration gradients between cathodes and anodes, thereby leading to serious decay of conductivities, which limits the discharging/charging rate, energy density and battery life.⁴ As the current widely-used polymer electrolytes for LIBs/LBs,

polymers such as poly(vinylidene fluoride) (PVDF), poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) or poly(ethylene oxide) (PEO) have been widely accepted as promising SPE candidates with suitable conductivity and processability in the past few years.⁵ But the general low t_{Li^+} value of these electrolytes has limited their applications. Although the use of single-ion conducting solid-state electrolytes based on ionic polymer matrices with immobilized counter-anions has been proven to be an effective approach to promote the preferential transport of Li^+ , the complex multistep synthesis and relatively low ionic conductivity of the electrolytes limit their scalable production and further application.⁶ The development of a universal approach to turn such conventional SPEs into practicable SPEs with quality remains a huge challenge.

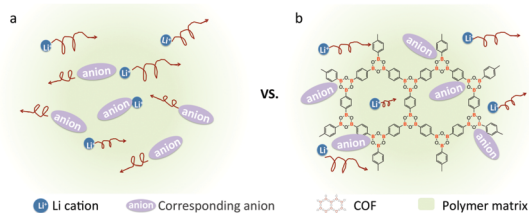
In recent years, some efforts have been devoted towards the development of functional additives for SPEs to form complexes of anions within the electrolytes, aiming at increasing the fraction of mobile Li^+ and thus improving the t_{Li^+} values.⁷ To realize the maximum advantages of these functional additives, two major challenges need to be addressed: the strong adsorption ability of the anions as well as the sufficient ionic adsorption sites within the additives. Several previous studies have reported the use of boron compounds as additives to increase the Li^+ transport ability of the electrolytes, considering that the boron atom is sp^2 -hybridized and contains an empty p-orbital that can interact with the basic anion of lithium salt.⁸ Currently, more attention is focused on developing boron-containing nanomaterials by tethering functional groups onto nanoparticles or solid films, with the aim to provide a high surface area that allows more effective adsorption of the free anions that will arise during cell operation.⁹ But to date, mainly due to the limited functional group density or relatively low active area of the additives that hinder the anion adsorption on additives, the real effect for enhancing the t_{Li^+} is still far from practical applications. As is well-known, covalent organic frameworks (COFs) are a novel class of crystalline solids that are characterized by their well-defined frameworks, high surface areas and permanent porosity.¹⁰ They have been widely investigated as capturers of carbon dioxide,

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/c8cc08725c

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Scheme 1 Schematic diagram representing the anion adsorption capacity of boron-containing COFs. (a) The case of SPEs without additives. Some Li^+ salt was dissociated into free ions, and both Li^+ and counter anions can diffuse in the polymer matrix. (b) The case of SPEs with COF additives. Strong adsorption of the anions by the boron group in the framework weakens the diffusion of the anions during battery operation, resulting in an enhanced t_{Li^+} .

catalytic materials, or ionic conductors, but never as functional additives for enhancing the Li^+ transference number of SPEs.¹¹ Compared with the reported boron-containing additives, porous and lightweight COFs seem to be an ideal candidate, as they have a tunable functional (boron-containing) framework with high porosity and surface area and thus provide abundant active sites for anion adsorption, which is not possible with most of the reported additives.

Herein, we have reported for the first time that COFs with boron-containing frameworks are excellent candidates for t_{Li^+} enhancement in SPEs. As shown in the schematic illustration in Scheme 1, the advantages include: (1) the boron-containing frameworks enable strong anion adsorption and thus yield more free mobile Li^+ ; (2) the high porosity and surface areas of COFs provide abundant anion adsorption sites in the structure. We demonstrated that SPEs with COF additives show a significantly enhanced t_{Li^+} compared with that without COFs. In order to further realize the maximum advantages of these additives, modified COFs with higher surface areas than highly crystalline COFs with a large size were synthesized *via* a simple template-free approach in this work. Our result showed that for a COF-containing SPE, the higher the surface areas of the COF additives, the higher the t_{Li^+} of the SPE that can be obtained, accompanied with a slight improvement of the ionic conductivity of the SPE. This should be a general structural design guideline for COF additives used in SPEs. Our COF additive-based approach has provided a combination of advantages of reliability, convenience, and low-cost with an elevated t_{Li^+} for practicable SPEs, considering that the porous COFs in this work are easily obtained, and the final COF-containing SPEs can also be prepared by a simple mixing process.

COF-1 $[(\text{C}_3\text{H}_2\text{BO})_6(\text{C}_9\text{H}_{12})_1]$, the first boron-containing COF synthesized by Yaghi's group, was chosen as the model COF in our work, considering that it can be efficiently synthesized in high yield under mild reaction conditions.¹² Different from the reported solvothermal method, here we adopted a template-free approach to synthesize COF-1 with different porosity (named H-COF-1 hereafter), which was previously developed in the synthesis of porous MOFs.¹³ The concept is shown as a scheme in Fig. 1a. The typical synthesis process can be found in the ESI.†

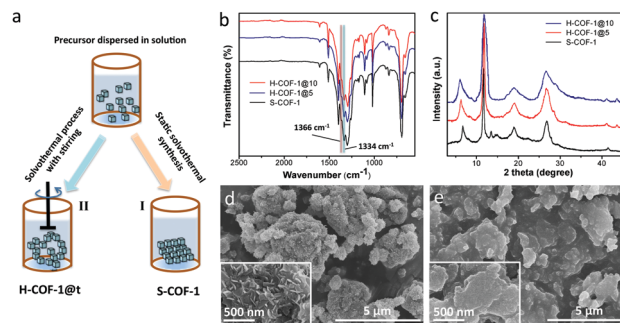


Fig. 1 (a) Different synthesis process of COF-1. I, S-COF-1 with a relatively ordered structure obtained *via* a static solvothermal method; II, H-COF-1@t with a porous structure fabricated under strong stirring. (b) FTIR spectra of different COF samples. The characteristic boron functional group bands of the COFs are highlighted. (c) X-ray diffraction analysis of different COF samples. (d) SEM images of H-COF-1@10 and (e) SEM images of S-COF-1.

Theoretically, three boronic acid molecules would converge to form a planar six-membered B_3O_3 (boroxine) ring with the elimination of three water molecules (Fig. S1, ESI†).¹² The Fourier transform infrared (FTIR) spectra shown in Fig. 1b indicate the formation of the expected boron-containing ring groups in all COF samples. The peaks at ~ 1334 and 1366 cm^{-1} correspond to the B–O stretching, which is characteristic of boroxine (Fig. 1b).¹² All the FTIR spectra matched well with the previous reports, confirming that the expected COFs had formed.¹² The powder X-ray diffraction (XRD) spectra of H-COF-1@t, especially H-COF-1@10, showed broader XRD peaks than that of S-COF-1, indicating a lower crystalline structure due to the stirring-assisted reaction conditions (Fig. 1c). The scanning electron microscopy (SEM) images of the samples showed markedly different morphologies. As can be seen in Fig. 1d and e, S-COF-1 exhibited a continuous layered structure, while H-COF-1@10 and H-COF-1@5 (for more details see Fig. S2, ESI†) have a close-to-spherical shape with diameters ranging from sub-microns to a few microns, which was much smaller than the solvothermally-prepared S-COF-1. More interestingly, compared to the relatively smooth surface of S-COF-1, both H-COF-1@10 and H-COF-1@5 exhibited a rough surface, where vast nanosheets with an average thicknesses of a few nm grow randomly, implying the rich porous structure of the H-COF-1@t samples.

S-COF-1 showed a surface area of $690 \text{ m}^2 \text{ g}^{-1}$ (Fig. S2 and Table S1, ESI†), which is close to the reported value.¹² But the H-COF-1@t samples showed significantly increased surface areas, in particular, H-COF-1@10 exhibited the highest value of $812 \text{ m}^2 \text{ g}^{-1}$. The increase in the reaction time (more than 10 h) did not further improve the surface area of H-COF-1@t.

A widely used PVDF-based gel polymer electrolyte was selected as the model substrate in the present study, considering that it has demonstrated sufficient conductivity at room temperature for commercial use in LIBs/LBs, which also provides an opportunity to study the effect of t_{Li^+} enhancement under battery operating conditions.¹⁴ PVDF composite electrolytes containing LiClO_4 and COF additives in various compositions were prepared by a solution casting technique (named PVDF/S-COF-1 and PVDF/H-COF-1@t, respectively). SEM and EDX analyses indicated an even distribution

of the COFs and Li salt in the resultant films (Fig. S4, ESI†). To maintain the focus on the t_{Li^+} of the electrolytes, PVDF-based films with high porosity were prepared for this study because they would provide higher ionic conductivity and also more active sites to maximize the potential of the COF additives for t_{Li^+} enhancement, although excessive porosity is a clear disadvantage for assembling an all solid-state cell as it may make the polymer film more brittle.

The ionic conductivity of the PVDF-based electrolytes with different COF additives is presented in Fig. 3a. Interestingly, all the COF-containing electrolytes show generally higher conductivity than pure PVDF electrolyte, possibly because of the strong anion-adsorption ability of the COF additives, thus promoting ion-pair dissociation within the electrolytes, which will be discussed later. The PVDF/H-COF-1@10 film exhibits the highest conductivity at room temperature ($\sim 2.5 \times 10^{-4} \text{ S cm}^{-1}$, with 20 wt% COF additives). The ionic conductivities of the COF-containing electrolytes decrease when the additive content exceeds 20 wt%, possibly due to the poor dispersion of the COFs that would disturb the ionic conduction path (Fig. S5, ESI†). The t_{Li^+} for all electrolytes was measured using the DC polarization/AC impedance combination method.^{3a} The electrolyte film was sandwiched between symmetric lithium foils and the current response was measured at a constant potential and the impedance spectra were collected. As shown in Fig. 2b and Table S2 (ESI†), the t_{Li^+} of the pure PVDF electrolyte is only 0.36, while PVDF/S-COF-1 and PVDF/H-COF-1@t exhibited higher t_{Li^+} and the value increased with the COF content. The highest t_{Li^+} values were obtained for the samples with 20 wt% COF additives. The maximum t_{Li^+} values of 0.46, 0.61 and 0.71 were obtained for PVDF/S-COF-1, PVDF/H-COF-1@5 and PVDF/H-COF-1@10 with 20 wt% COF content, respectively. All the data above confirmed the COF surface area dependent conductivity and t_{Li^+} properties of the COF-containing electrolytes. H-COF-1@10 with the largest surface area provides the highest ionic conductivity as well as an improved t_{Li^+} value.

The enhanced t_{Li^+} values are mainly attributed to the strong anion adsorption of the boron-based groups, which have been reported in previous studies.^{9b} The polarizable anion ClO_4^- is expected to interact with one or more B atoms in the frameworks to yield a complex. FTIR spectra of the different samples are thus compared because the FTIR signals at the 600–650 cm^{-1} range related to the ClO_4^- anions are highly sensitive to the ion association status.¹⁵ Two characteristic sharp peaks at ~ 626 and

635 cm^{-1} due to the vibration modes of the ClO_4^- ions were observed for all samples. The peak at $\sim 626 \text{ cm}^{-1}$ was assigned to the vibration of the “free” ClO_4^- anion, which does not interact directly with the Li^+ cations, and the peak centered at $\sim 635 \text{ cm}^{-1}$ was assigned to the vibration of the LiClO_4 pairs.¹⁵ On the basis of the integrated peak areas, an estimated only $\sim 21\%$ of the ClO_4^- anions are present in the form of aggregated ion pairs for the PVDF/H-COF-1@10 sample, while the sequential increase of “aggregated” ClO_4^- anions was observed in the PVDF/H-COF-1@5, PVDF/S-COF-1 and PVDF electrolytes, respectively (Fig. S6, ESI†). These results suggested that COF-1 could be an anion receptor and can lead to an additional dissociation of LiClO_4 by Lewis acid–Lewis base interaction between the boron-based groups and the ClO_4^- anions, but the precise mechanism of this interaction in COF-1 is uncertain at present. On the other hand, the extensive interaction of the anions and the boron-containing framework may, in turn, promote the Li salt dissociation (generating more free Li^+), thus resulting in the increased ion conductivity of the COF-containing electrolytes.¹⁶

The enhanced conductivity and t_{Li^+} of the SPEs benefit the building of better solid-state cells. The electrochemical properties of the electrolytes were firstly studied in symmetric $\text{Li}||\text{electrolyte}||\text{Li}$ cells using a constant-current polarization technique. The electrolytes with 20 wt% COF additives were used for studies. The polarization potentials of the $\text{Li}||\text{electrolyte}||\text{Li}$ cells were measured as a function of the current density. Fig. 3a and Fig. S7 (ESI†) show the results in the current density range of 0.05–1 mA cm^{-2} at room temperature. The cells with the PVDF, PVDF/S-COF-1, PVDF/H-COF-1@5 and PVDF/H-COF-1@10 electrolytes exhibited polarization curves with overpotentials of ~ 9.2 , 6.8, 5.9 and 4.5 mV at a low current density of 0.05 mA cm^{-2} , respectively. All the cells exhibited increasing polarization potentials with increasing current density, while the cells with the COF-containing electrolytes exhibited lower polarization potentials compared to the $\text{Li}||\text{PVDF}||\text{Li}$ cell operated at the same current density. As expected, the $\text{Li}||\text{PVDF/H-COF-1@10}||\text{Li}$ cell shows the lowest and stable polarization potentials at different current densities (Fig. 3a and Fig. S7, ESI†). As is well known, a low t_{Li^+} will lead to the formation of a concentration gradient with deleterious effects such as lithium dendritic growth. Due to the increased fraction of mobile Li^+ in the overall ionic current of the COF-containing electrolytes, the concentration gradient and polarization during cell operation would be significantly suppressed, thereby increasing the electrochemical efficiency of the cells. The linear sweep voltammetry result of the PVDF/COFs film suggests that it can be employed in conventional lithium ion battery systems (Fig. S8, ESI†). The solid-state cells $\text{Li}||\text{PVDF}||\text{LiFePO}_4$ and $\text{Li}||\text{PVDF/H-COF-1@10}||\text{LiFePO}_4$ were thus assembled, respectively. The charge–discharge curves (at a current density of 1C) of the cells at room temperature are shown in Fig. 3b. The $\text{Li}||\text{PVDF/H-COF-1@10}||\text{LiFePO}_4$ cell exhibited a decreased overpotential and stable cyclic behavior. It demonstrated an initial capacity of 128 mA h g^{-1} , and the capacity retention ratio is 94% after 80 cycles (Fig. S8, ESI†). Most remarkably, the $\text{Li}||\text{PVDF/H-COF-1@10}||\text{LiFePO}_4$ cell also showed enhanced rate performance with discharge capacities of ~ 145 , 138, 128, 122

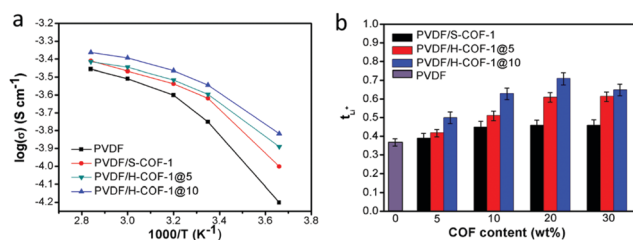


Fig. 2 (a) Temperature dependence of the ionic conductivity of different electrolytes (with 20 wt% COF additives). (b) t_{Li^+} of the electrolytes with different COF contents.

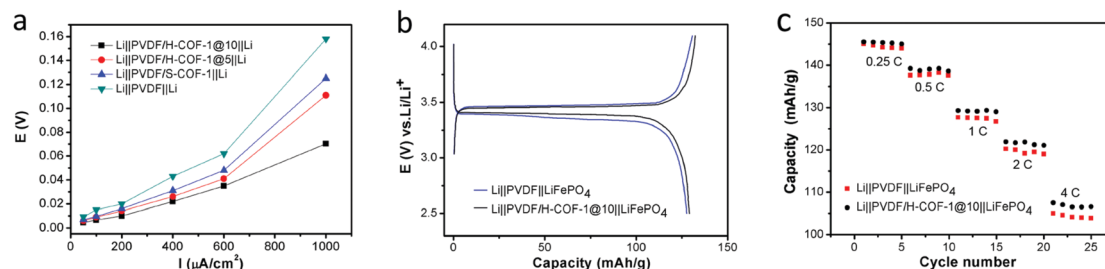


Fig. 3 (a) Curves of constant current polarization for the Li||electrolyte||Li cells. (b) Charge–discharge curves of the Li||PVDF||LiFePO₄ and Li||PVDF/H-COF-1@10||LiFePO₄ cells at room temperature. (c) Rate performances of the Li||PVDF||LiFePO₄ and Li||PVDF/H-COF-1@10||LiFePO₄ cells at room temperature.

and 107 mA h g^{−1} at 0.25C, 0.5C, 1C, 2C and 4C rates, respectively (Fig. 3c). The EIS curve of the cell is shown in Fig. S10 (ESI[†]).

In summary, porous COFs with boron-containing frameworks have been demonstrated as efficient additives for Li⁺ transference number enhancement in SPEs. The boron-containing frameworks enable strong anion adsorption capacity and the porous frameworks also provide abundant active sites for ion adsorption. A series of PVDF electrolytes with both high Li⁺ transference number and high ion conductivity can be simply prepared by incorporation of porous COFs into a polymer host. The solid-state cells assembled with the COF-containing SPEs exhibited remarkably decreased overpotentials and enhanced rate performances. Such COF-additive approaches are convenient and economical for next-generation solid-state batteries with enhanced energy density and safety, although continued improvement in COF synthesis and understanding of the fundamental mechanism are required in subsequent research. The nature and geometry of the boron-containing COFs determine the degree of anion adsorption, which in turn determines the Li⁺ transference number. Incorporation of electron-withdrawing groups such as fluorinated aryl and alkyl groups into the frameworks may represent a promising approach for the further enhancement of the Li⁺ transference number, which can be expected considering the almost unlimited possibility in the molecular design and synthesis of COF materials.

This work was supported by the National Natural Science Foundation of China (Grant No. 21703072). H. C. acknowledges the support from the Promotion Program for Young and Middle-aged Teacher in Science and Technology Research of Huaqiao University. D. D. acknowledges the Subsidized Project for Post-graduates' Innovative Fund in Scientific Research of Huaqiao University.

Conflicts of interest

The authors declare no competing conflicts of interest.

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